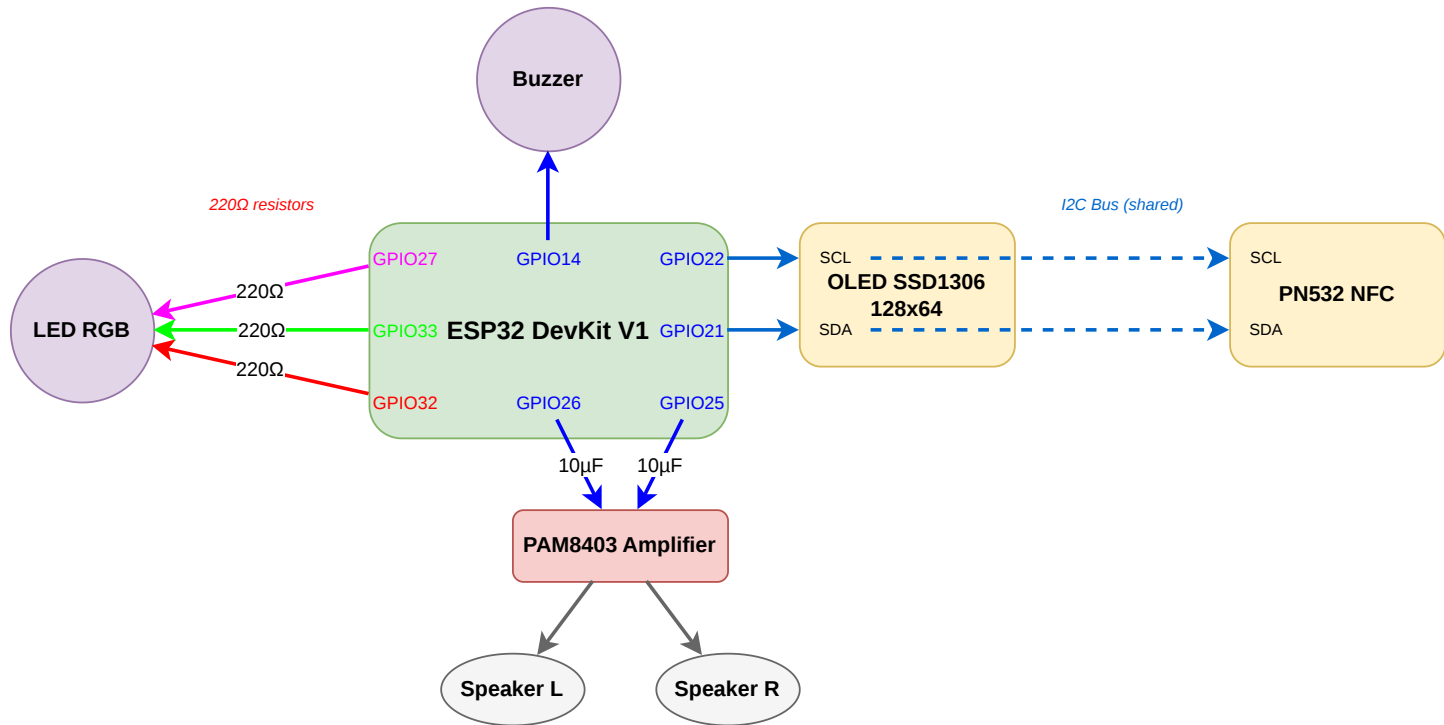




TapTune - Hardware Connection Table

Component	Pin / Name	Connected To	Notes
ESP32 DevKit V1	GPIO21	OLED SDA + PN532 SDA	Shared I2C bus
	GPIO22	OLED SCL + PN532 SCL	Shared I2C bus
	GPIO32	LED Red (220 Ohm)	Common cathode -> GND
	GPIO33	LED Green (220 Ohm)	Common cathode -> GND
	GPIO27	LED Blue (220 Ohm)	Common cathode -> GND
	GPIO14	Buzzer (+)	Passive buzzer, PWM
	GPIO25	PAM8403 IN L (10uF)	Audio left channel
	GPIO26	PAM8403 IN R (10uF)	Audio right channel
	3.3V	OLED VCC + PN532 VCC	Power for I2C devices
	GND	OLED GND + PN532 GND*	Common ground
OLED SSD1306	SDA	ESP32 GPIO21	I2C data
	SCL	ESP32 GPIO22	I2C clock
	VCC	ESP32 3.3V	Power
	GND	ESP32 GND	Ground
PN532 NFC	SDA	ESP32 GPIO21	I2C data
	SCL	ESP32 GPIO22	I2C clock
	VCC	ESP32 3.3V	Power
	GND	ESP32 GND	Ground
LED RGB	R	ESP32 GPIO32	220 Ohm resistor
	G	ESP32 GPIO33	220 Ohm resistor
	B	ESP32 GPIO27	220 Ohm resistor
	Catodo	GND	Common cathode
Buzzer passivo	+	ESP32 GPIO14	PWM LEDC
	-	GND	
PAM8403	IN L	ESP32 GPIO25	10uF capacitor
	IN R	ESP32 GPIO26	10uF capacitor
	VCC	5V (USB or external)	Amplifier power
	GND	GND	
	OUT L	Speaker Left	
	OUT R	Speaker Right	
Speaker L	+ / -	PAM8403 OUT L	
Speaker R	+ / -	PAM8403 OUT R	

Notes: I2C bus operates at 400 kHz. PAM8403 requires separate 5V supply. PN532 DIP switches: SW1=ON, SW2=OFF. *OLED GND + PN532 GND + LED + Buzzer + PAM